Games in the real world
(focus on energy-related applications)

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Game Theory and Mechanism Design - Lecture 15

Learning objectives

At the end of this lecture (and related tutorial(s)), a student should be able to:

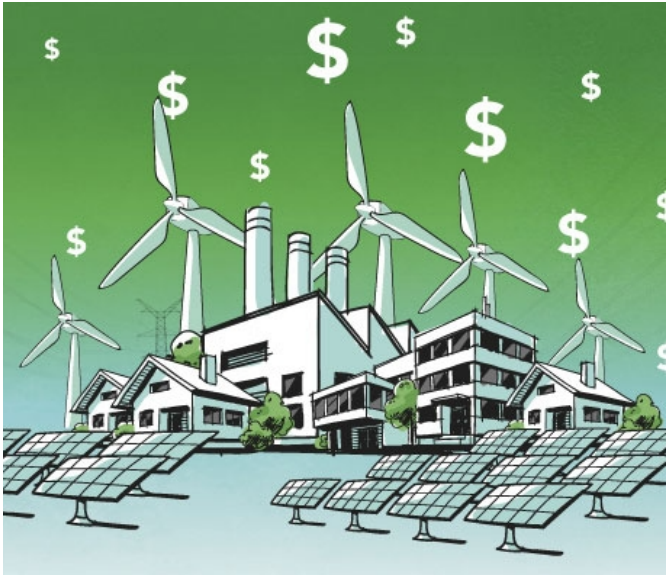
- See how we can **recognise games** in real-world problems
- Use **acquired knowledge** to analyse these games
- Appraise potential **limitations**

Outline

1. Game 1: The electricity market
2. Game 2: Incentives in electricity markets
3. Game 3: Minority games in electricity markets
4. Game 4: Monetizing data in ML/analytics problems

1. **Game 1: The electricity market**

The (wholesale) electricity market



[source: fraserinstitute.org]

- Electricity markets are there as an interface between electricity producers (power plants, wind farms, etc.) and consumers
- As for any market, there are game-theoretical aspects and a mechanism design component involved...

Basics of the game

It is a 2-sided (resource allocation) game

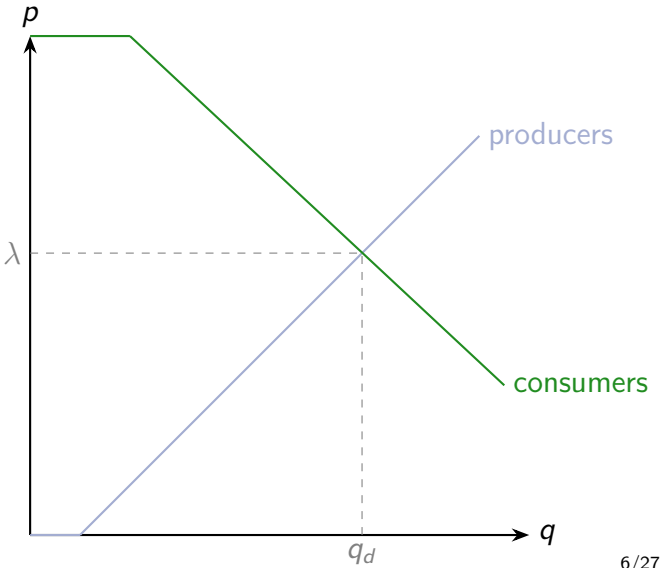
They are ranked and "matched" based on a *social welfare maximisation principle*

Producers

Producers (set $N_p = \{p_1, \ldots, p_m\}$) place quantity-price offers, i.e., (q_i, λ_i) for producer i

Consumers

Consumers (set $N_c = \{c_1, \ldots, c_n\}$) place quantity-price offers, i.e., (q_j, λ_j) for consumer j



Outcome of the game

Everybody on the left of the equilibrium point is scheduled

Definition (resource allocation)

For a set of producers N_p and a set of consumers N_c, leading to the equilibrium point (q_d, λ), the outcome in terms of *resource allocation* is such that

- all producers offering prices $\lambda_i \leq \lambda$ are scheduled. Others are not
- all consumers offering prices $\lambda_j \geq \lambda$ are scheduled. Others are not

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But, to define pay-offs, we need both **quantity** and **prices**.

Definition (marginal pricing)

For a set of producers N_p and a set of consumers N_c, leading to the equilibrium point (q_d, λ), the outcome in terms of *marginal pricing* is such that

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- all scheduled consumers pay the price λ

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Definition (pay-as-bid pricing)

For a set of producers N_p and a set of consumers N_c, leading to the equilibrium point (q_d, λ), the outcome in terms of *pay-as-bid pricing* is such that

- all scheduled producers are paid at price λ_i they offered
- all scheduled consumers pay the price λ_j they offered

Let's play

Here is a simplified version of the game:

- you each have 1MWh you want to sell (so, quantity fixed)
- it costs you between £0 and £5 (pick a number, and write it down)
- we know the consumption to meet is 5MWh, and they are ready to pay up to £10/MWh

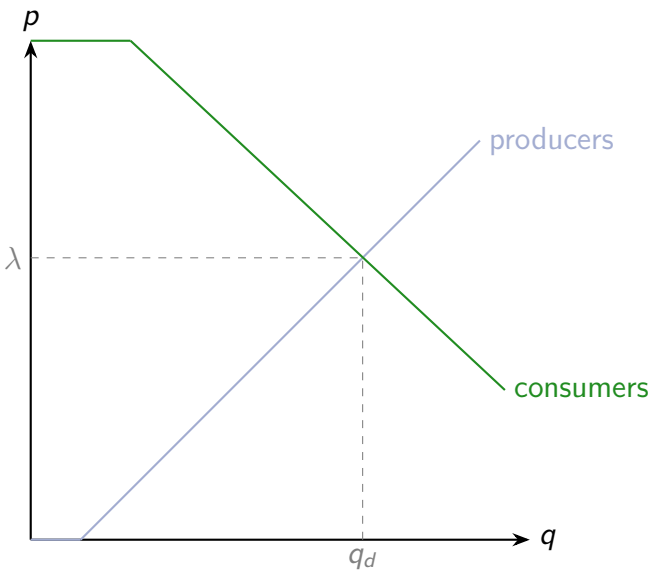
We will play the game with the two pricing approaches...

Pay-as-bid pricing

Marginal pricing

2. **Game 2: Incentives in electricity markets**

The same game... nearly.



Consider the same basic setup in terms of the electricity market. However,

- we allow price offers in a range between £-100 and £500
- for renewable energy producers (wind, solar), their marginal cost is £0
- we want to support them... so, whenever they are scheduled, we promise them subsidies (£30/MWh)

What is their best-response strategy in the market?

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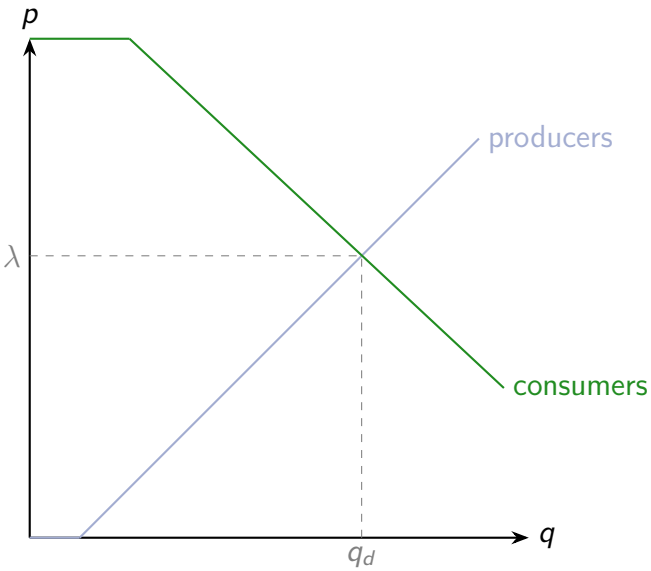

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What is their best-response strategy in the market?

Now, I say that when the market is cleared, if the overall price is negative, they will lose their subsidy... what is then their best-response strategy? (there may be several options...)

3. **Game 3: Minority games in electricity markets**

Minority games

Minority games are about being on the right side...

The game:

- A population with N individuals (e.g., $N = 101$)
- They may want to go to the bar to attend a concert
- Capacity of the bar is m people (e.g., $m = 60$)
- n people eventually show up ($n \in [0, N]$)

The pay-off function for player i is

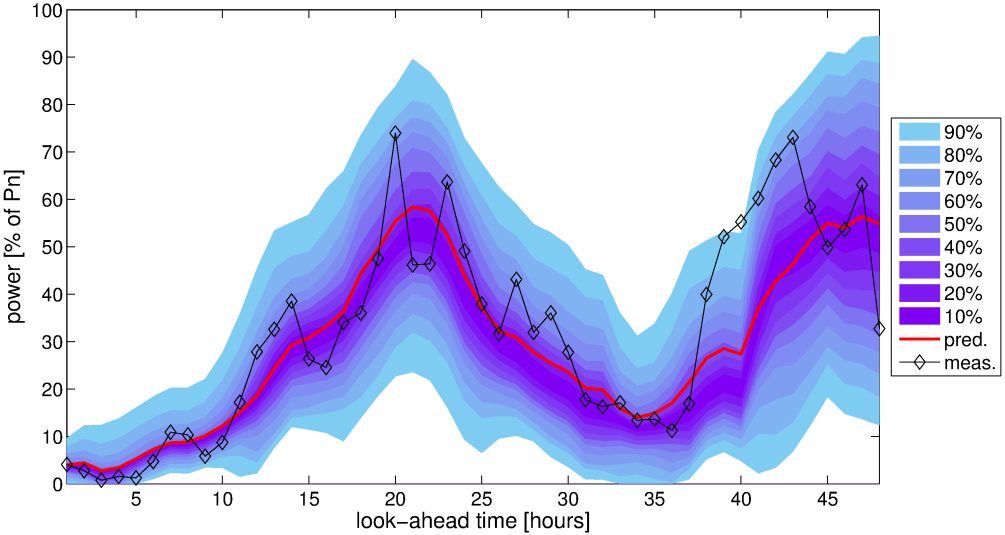
$$s_i = \begin{cases} 1, & \text{if going, and if } n \leq m \\ -1, & \text{if going, and if } n > m \\ 0, & \text{if staying home} \end{cases}$$

Do you see such games in the real world?

And, what is to happen?

Renewable energy forecasts are uncertain

When trading in electricity markets, the best is to have probabilistic forecasts for future power generation (trading is a noon, for the following day, 00:00 to 00:00)



Penalties

You can be wrong, but should be on the right side!

Produce more than scheduled

- if the overall production is more than consumption, you get penalised
- if the overall production is less than consumption, you get rewarded

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So, when you trade... you basically **wish you were wrong**, as long as it is in the right direction.

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Proposition (Optimal offering strategies)

In the described market, your optimal offering strategy (as a renewable energy producer, with uncertain production) is **independent** *of your forecast for future energy generation(!). Instead,*

- *one predicts the sign η of the imbalance in the system (i.e., $\eta > 0$ for overall production is more than consumption)*
- *if one predicts $\eta > 0$, you offer 0*
- *if one predicts $\eta < 0$, you offer your maximum production*

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What do you think would happen if all renewable energy producers do that??

The population effect

What a single market participant does may not be an issue...

Definition (price-taker)

A single renewable energy producer considers themselves as a *price-taker* if their decision do not affect market outcomes.

But, if they all do that, this will have an impact!

Definition (Population price-maker effect)

Even though a single renewable energy can be seen as a price-taker on their own, they may become *price-maker* as a group... I.e., the aggregation of their decision will impact market outcomes.

Then, as a single renewable energy producer, what should be your strategy?

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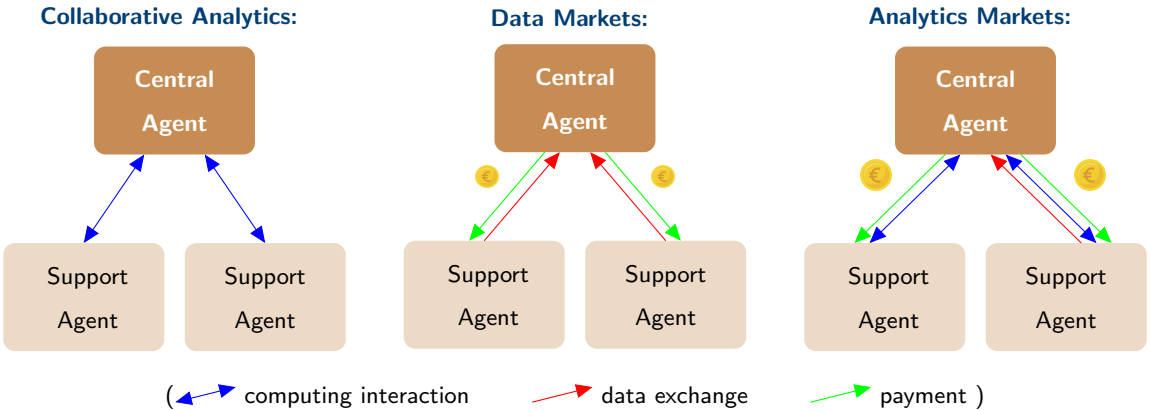
... and this is why I started looking at minority games...

4. **Game 4: Monetizing data in ML/analytics problems**

Analytics platforms

Agents meet through **analytics platforms** supporting collaborative and market-based analytics



The central agent and the regression problem

- Consider a *central agent* ("**Forecaster**") with a regression problem, e.g., as a basis to forecast renewable power generation for a given site (y_{t+k})
- **Forecaster** owns a set ω of m features, $\omega = \{x_1, \ldots, x_m\}$

The following regression problem could be used as basis for eventual prediction,

$$Y_{t+k} = \beta_0 + \sum_{i=1}^{m} \beta_i x_{i,t} + \varepsilon_t, \quad t = 1, \ldots, T$$

The vector of parameters $\boldsymbol{\beta} = [\beta_0 \ \ldots \ \beta_m]^\top$ can easily be learned by minimizing an appropriate loss function

$$\hat{\boldsymbol{\beta}} = \underset{\boldsymbol{\beta}}{\operatorname{argmin}}\, S_\omega(\boldsymbol{\beta}), \qquad S_\omega(\boldsymbol{\beta}) = \frac{1}{T} \sum_{t=1}^{T} \rho \left(y_{t+k} - (\beta_0 + \sum_{i=1}^{m} \beta_i x_{i,t}) \right)$$

where ρ may be any convex loss function (e.g., quadratic, pinball loss, etc.)

Based on the data available, the minimum loss function value is $S^*_\omega = S_\omega(\hat{\boldsymbol{\beta}})$

- **Forecaster** could post the regression task on an analytics platform, to improve model fit
- **Forecaster** declares a willingness to pay of ϕ =1€ per percent-point improvement in S and per data point provided.

Support agents and the augmented regression problem

- Two *support agents* **Good Data** and **Useful Features** may bring in additional features z_1 and z_2, to be remunerated
- The overall set of features now is $\Omega = \omega \cup \{z_1, z_2\}$

The regression problem can then be augmented, as

$$Y_{t+k} = \underbrace{\beta_0 + \sum_{i=1}^{m} \beta_i x_{i,t}}_{\text{Forecaster}} + \underbrace{\gamma_1 z_{1,t}}_{\text{Good Data}} + \underbrace{\gamma_2 z_{2,t}}_{\text{Useful Features}} + \varepsilon_t, \quad t = 1, \ldots, T$$

where the augmented vector of coefficients $\boldsymbol{\beta}^+ = [\beta_0 \ \ldots \ \beta_m \, \gamma_1 \, \gamma_2]^\top$ can be learned similarly, by minimizing an appropriately chosen convex loss function ρ, i.e.,

$$\hat{\boldsymbol{\beta}}^+ = \underset{\boldsymbol{\beta}^+}{\operatorname{argmin}} \, S_\Omega(\boldsymbol{\beta}^+), \qquad S_\Omega(\boldsymbol{\beta}^+) = \frac{1}{T} \sum_{t=1}^{T} \rho \left(y_{t+k} - (\beta_0 + \sum_{i=1}^{m} \beta_i x_{i,t} + \gamma_1 z_{1,t} + \gamma_2 z_{2,t}) \right)$$

We eventually write $S_\Omega^* = S_\Omega(\hat{\boldsymbol{\beta}}^+)$

- If z_1 and/or z_2 are informative features, one expects $S_\Omega^* < S_\omega^*$

Revenues, payments and market properties

- How to define **revenues** and **payments** in such a regression market?

For **Forecaster**, the payment is

$$\pi_c = \phi(S^*_\omega - S^*_\Omega)\, T$$

For each support agent j ($j = 1, 2$), the revenue is given by

$$\pi_j = (S^*_\omega - S^*_\Omega)\, T\, \phi\, \psi_j, \quad j = 1, 2$$

where ψ_j is an allocation policy based on feature valuation (can be obtained with, e.g., leave-one-out or Shapley-based allocation), such that $\sum_j \psi_j = 1$

Such a simple approach actually yields a market with a wealth of good properties, i.e.,

- budget balance
- symmetry (or anonymity)
- zero element
- incentive compatibility (truthfulness)
- individual rationality

In-sample and out-of-sample

Typically,

- we learn in-sample (batch or online)
- we predict out-of-sample...

Can we use the above concepts more generally?

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Can we use the above concepts more generally?

In **online regression** markets, the payments can be generalized with

$$\pi_{j,t} = (S^*_{\omega,t} - S^*_{\Omega,t})\, \phi\, \psi_{j,t}, \quad j = 1, 2$$

where $S^*_{\omega,t}$ and $S^*_{\Omega,t}$ are time-varying estimator of the loss function, and $\psi_{j,t}$ is a time-varying estimate of allocation policies (profiting of their linearity property)

And, in **out-of-sample regression** markets (i.e., for genuine forecasting),

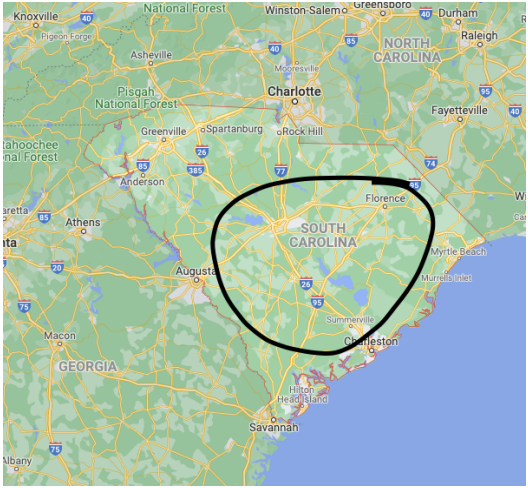
$$\pi_{j,t} = (s^*_{\omega,t} - s^*_{\Omega,t})\, \phi\, \psi_{j,t}, \quad j = 1, 2$$

where $s^*_{\omega,t}$ and $s^*_{\Omega,t}$ are time-varying estimator of the loss function, and $\psi_{j,t}$ is the instantaneous allocation policies (i.e., readily Shapley additive explanation)

Batch, online, and out-of-sample regression markets all enjoy the same properties.
(given convex loss functions and models that are linear in their parameters)

A real world case study in South Carolina (USA)

Wind power generation for 9 locations in South Carolina (US) – 7 years of data with hourly resolution



Agent	P_n [MW]	Lat./Long.	County
a_1	1.75	34.248/-79.75	Florence
a_2	2.96	34.02/-79.537	Florence
a_3	3.38	33.925/-79.958	Florence
a_4	16.11	34.732/-82.122	Laurens
a_5	37.98	34.556/-81.889	Laurens
a_6	30.06	34.334/-82.133	Laurens
a_7	2.53	33.136/-80.857	Colleton
a_7	2.6	33.112/-80.665	Colleton
a_9	1.24	32.641/-80.504	Colleton

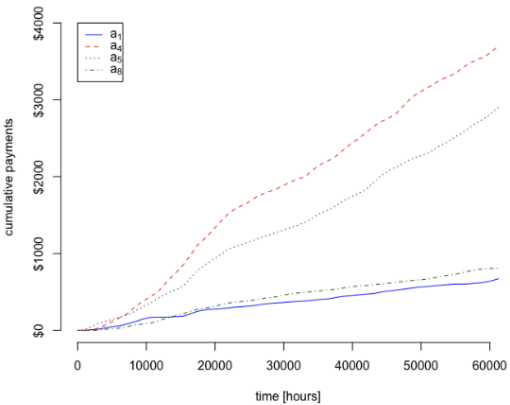
1-hour ahead forecasting based on $Y_{i,t} = \beta_0 + \sum_{\delta=1}^{\Delta} y_{i,t-\delta} + \sum_{j \neq i} \sum_{\delta=1}^{\Delta} y_{j,t-\delta} + \varepsilon_{i,t}$

where a_i is the central agent and a_j $(j \neq i)$ are the support agents

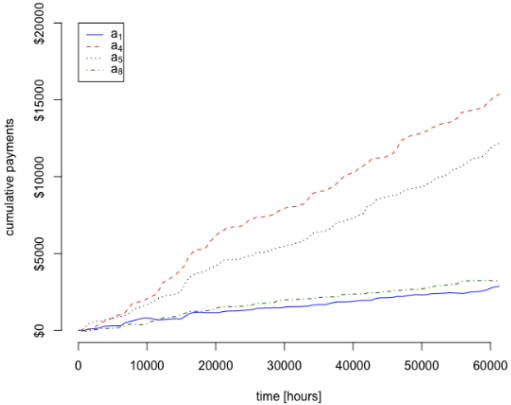
Online and out-of sample regression markets

- Online quantile regression ($\tau = 0.55$) in models with 2 lags for a_i and 1 lag for a_j ($j \neq i$)
- $\phi = 0.2$\$ in-sample, and $\phi = 0.8$\$ out-of-sample (per unit loss, per data point)

Cumulative payments of a_6 towards others:



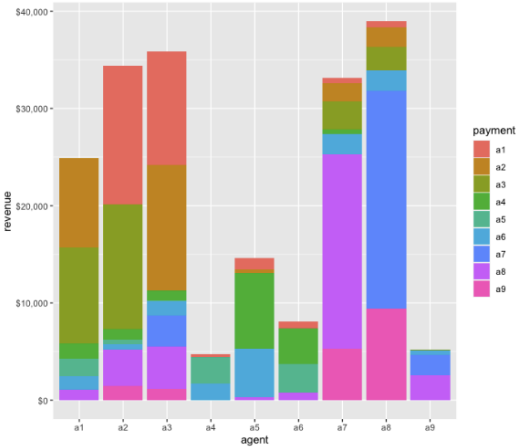
(a) Online regression market.



(b) Out-of-sample regression market.

Overall payments

Let's see what happens if they all pay each other for data to improve forecasts...



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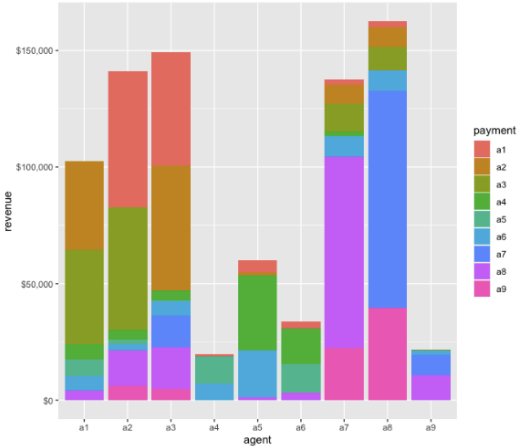

(b) Out-of-sample regression market.

On a per-data-point basis:

- a_4 only gets 0.39$ per data point
- a_8 gets 3.26$ per data point

Wrapping up

- There are so many games in **energy-related applications**
- Games are also a way to look at how **people interact with technology**
- Games and mechanisms are also ways to **coordinate** agents towards a desirable social outcome.
- Games with **data** comprise an increasingly exciting topic!

Thanks for your attention!